Subaita Mamun

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Education

University of Alberta — Expected April 2028
B.Sc. Computer Science, Honors in Artificial Intelligence — *Edmonton, AB*

Technical Experience

Applied Machine Learning Scientist (Incoming) — January 2027 – April 2027
TD Bank Group, AI2 Wealth — *Toronto, ON*
- Will contribute to the AI2 Wealth Generative AI team's model development and evaluation workflows, supporting the assessment of model performance and reliability for business applications.

Data Science Intern — September 2026 – December 2026
TD Bank Group, AI2 Wealth — *Toronto, ON*
- Enabled systematic evaluation of agentic AI performance by developing an end-to-end pipeline using an LLM-as-a-Judge framework to assess response accuracy and calculate evaluation metrics.
- Reduced reliance on manual response review by 40% by engineering evaluation prompts and orchestrating specialized subagents to assess agent outputs against expected responses, producing consistent and repeatable assessments.
- Strengthened evaluation pipeline reliability by integrating authenticated LLM API endpoints and leveraging GitHub CLI to execute evaluation workflows.
- Translated business requirements into evaluation criteria and pipeline improvements by collaborating with stakeholders to align assessments with business needs.

Lead Data Engineer — September 2025 – Present
NAT-EHT Platform, Network for Applied Technology — *Edmonton, AB*
- Enabled reliable downstream analysis by building modular Databricks ETL pipelines to integrate, transform, and validate multi-source data.
- Standardized data from 30+ annual reports into relational models and developed Power BI dashboards to communicate trends and insights across 4+ nonprofit organizations.

Data Science Intern — January 2026 – April 2026
Canadian Museum of Nature — *Ottawa, ON*
- Made unstructured image collections searchable for curator research by building a semantic search application using CLIP embeddings and FAISS vector search.
- Achieved 90% F1 on a 14,000+ image classification task by training a CNN and evaluating augmentation and oversampling strategies to address class imbalance.
- Streamlined manual loan-request processing through Power Automate, Excel, and VBA automation; gathered workflow requirements from 7 domain experts and presented findings to 70+ employees.

Machine Learning Fellow — May 2026 – June 2026
AI4Good Lab, Alberta Machine Intelligence Institute — *Edmonton, AB*
- Automated household property inventory from disaster images by developing a YOLO-World detection pipeline and integrating Gemini API outputs into structured JSON inventories.
- Achieved 81% accuracy in distinguishing intact from partially damaged furniture by tuning model parameters and confidence thresholds; incorporated requirements from insurance brokers and disaster victims.

Projects

StockSense | *Python, FinBERT, XGBoost, MLflow, GitHub Actions* — October 2026
- Combined financial news, analyst ratings, and stock prices into a forecasting pipeline by engineering FinBERT sentiment and market features.
- Developed an XGBoost model to predict unusually large five-day stock movements, using walk-forward validation, MLflow experiment tracking, and model quality gates to evaluate performance against baselines.

SOILution | *1st Place, natHacks 2025 EcoTech Stream; 250+ teams* — November 2025
- Projected a 52% reduction in water consumption by developing a time-series forecasting solution that used sensor data and feature engineering to predict irrigation needs.

Technical Skills

Programming & Data: Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, ETL, data modeling
AI/ML: PyTorch, scikit-learn, XGBoost, NLP, Generative AI, LLMs, Agentic AI, LLM-as-a-Judge, prompt engineering, model evaluation, CLIP, FAISS
Platforms & Tools: Azure Databricks, GCP, MLflow, Airflow, Docker, Git/GitHub, GitHub CLI, REST APIs, CI/CD, Power BI, Excel, VBA, Gemini API